

Real-time Capable Learning-based Visual Tool Pose Correction via Differentiable Digital Twin

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Abstract—Autonomy in robot-assisted minimally invasive surgery has the potential to reduce surgeon cognitive and task load, thereby increasing procedural efficiency. However, implementing accurate autonomous control can be difficult due to poor end-effector proprioception. Joint encoder readings are typically inaccurate due to kinematic non-idealities in their cable-driven transmissions. Vision-based pose estimation approaches are highly effective, but lack real-time capability, generalizability, or can be hard to train.

In this work, we demonstrate a real-time capable, Vision Transformer-based pose estimation approach that is trained using end-to-end differentiable kinematics and rendering. We demonstrate the potential of this approach to correct for noisy pose estimates through a real robot dataset and the potential real-time processing ability. Our finetuned model reaches the same performance level as an existing optimization-based method. Our approach is capable of near real-time inference at 22 Hz.

I. INTRODUCTION

Robot-assisted Minimally Invasive Surgery (RMIS) platforms require accurate real-time end-effector pose estimation for autonomous policies, controllers, and emerging mixed reality interfaces in which pose information is presented through a digital twin [1], [2]. In RMIS systems such as the da Vinci Research Kit (dVRK) [3], joint-encoder errors can result in inaccurate state inputs and tool pose estimates.

A significant source of pose error stems from cable-driven transmissions, which can produce inaccurate joint angle readings due to friction, cable stretch [4], [5], and backlash from external forces applied to the tool shaft [6]. This makes computing an accurate hand-eye transform between the end-effector and the stereo endoscope origin difficult without external sensing.

Visual servo approaches mitigate forward-kinematic uncertainty by estimating the end-effector pose in the camera frame. Optical markers or RGBD sensors can help, but augmenting the camera or end-effector constrains applicability to real surgery due to biocompatibility, sterilizability, and miniaturization requirements. Instead, image-based approaches have used neural networks such as DeepLabCut (DLC) [7] for few-shot markerless keypoint and feature detection [8], [9]. Mask-based approaches leveraging pretrained segmentation models such as Segment Anything Model 2 (SAM 2) [10] reduce manual keypoint labeling and mitigate keypoint ambiguity during tool rotations [11], [12].

However, unlike with markers, image-based approaches must track articulated instruments with partially visible kine-

matic chains and align these features with current estimates in the camera’s image space (see Fig. 1). Because this alignment depends on projection geometry, the resulting estimation problem is non-convex and numerically sensitive, and is often handled using iterative vision-based approaches [8], [13].

By allowing gradients to be backpropagated through the renderer, Differentiable Rendering (DR) provides a direct way to align rendered robot geometry with image-space observations. Prior work has used DR or related rendering-based methods to reconstruct robot or surgical tool pose by iteratively minimizing silhouette or geometric losses [11], [12], [14]. One significant drawback of these methods is that they require multiple optimization or search steps to converge, which limits their applicability in real-time control.

In this abstract, we present a learning-based pose correction framework built around a differentiable digital twin of the dVRK. The digital twin provides end-to-end differentiable kinematics and rendering for surgical tool geometry, while the learned model performs single-shot correction from stereo image features and noisy proprioceptive predictions.

II. METHOD

A. Differentiable Digital Twin

We constructed a differentiable digital twin of the dVRK to connect the robot’s proprioceptive state with its expected visual appearance. The digital twin mirrors the kinematic structure of the Patient Side Manipulator (PSM) and Endoscopic Camera Manipulator (ECM), using reported joint angles and camera-to-robot base transforms to instantiate the corresponding virtual robot configuration. This provides a

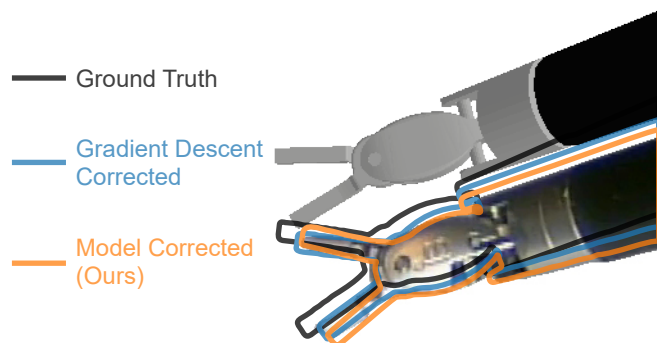


Fig. 1. Pose correction of a Large Needle Driver. The colored outlines show tool pose estimates projected onto the endoscopic image plane. The image pair compares the uncorrected digital twin rendering with the endoscopic reference image.

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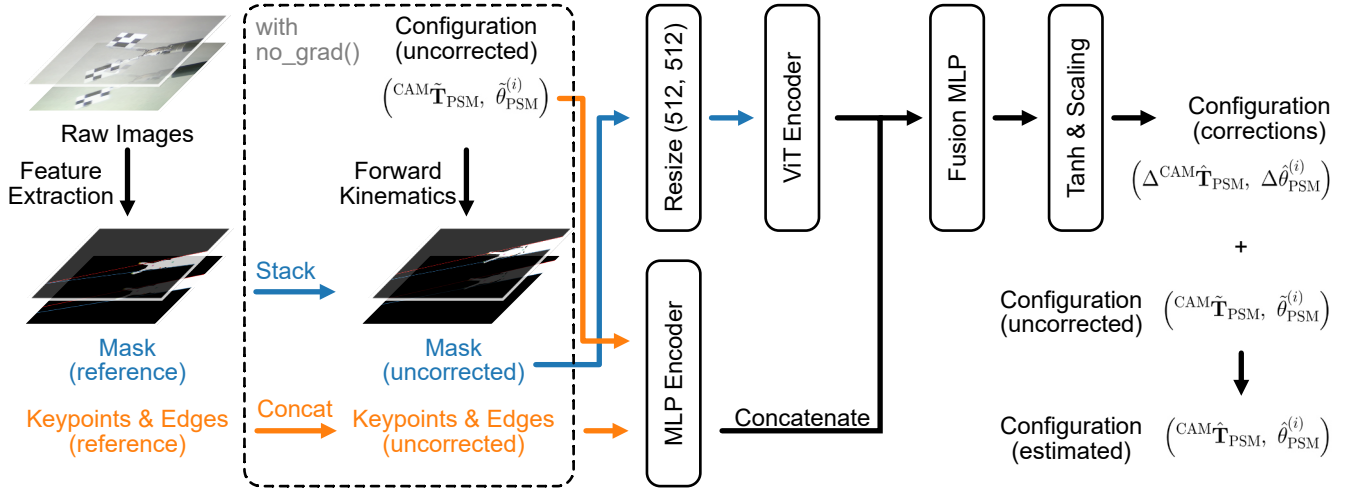


Fig. 2. Overview of the proposed vision-based deep learning pose estimation architecture.

synchronized geometric counterpart to the physical system, in which noisy encoder-derived states can be projected into the same endoscopic image space as the real observations.

The digital twin was implemented in Python using PyTorch Kinematics [15] and PyTorch3D [16]. It adopts the same APIs as the SurRoL [17] dVRK simulator and uses the same CAD models [3], preserving compatibility with existing surgical robot simulation workflows. Given a candidate robot configuration, the twin produces differentiable renderings of the tool silhouette, projected keypoints, and shaft edge lines. A hard Phong shader and a soft silhouette shader [18] are used to render endoscopic appearance and provide gradients for image-space alignment.

B. Model Architecture

We used a Vision Transformer (ViT) [19] to encode the visual inputs with two pairs of stereo masks being stacked in the channel dimension. As shown in Fig. 2, the visual inputs to the neural network are tool segmentation masks $\mathbb{M}_{\text{ref}}$ from the stereo endoscopic images, and uncorrected prediction stereo masks $\tilde{\mathbb{M}}$. In a real scenario, the former corresponds to PSM pose observed under the endoscope, while the latter corresponds to the robot’s noisy proprioception, and is derived from the joint encoder and a priori camera-to-robot base transform. This estimated pose is rendered by the digital twin.

Due to the partial observability of the kinematic chain, only the last four joints of the PSM (Outer Roll, Wrist Pitch, Wrist Yaw, and End Effector) are visible in the endoscopic view and can be corrected directly. Under this problem formulation, we adopt a similar parametrization as in prior works [8], [12], lumping the transform from the first observable joint of the PSM chain to the camera origin into a single term. This transform from the camera to PSM Outer Roll frame is parametrized as ${}^{\text{CAM}}\mathbf{T}_{\text{PSM}} \in \text{SE}(3)$.

The pipeline is modular: segmentation, keypoint detection, and shaft-line extraction are treated as interchangeable perception submodules rather than fixed components of the

pose correction network. This allows each subtask to use the most suitable available model, and future improvements in upstream perception models may further improve pose correction accuracy without changing the digital twin or correction architecture.

Keypoints $\mathbf{p}^{(i)}$ and edge lines $\mathbf{l}^{(i)}$ are extracted from the stereo endoscopic images, and the proprioception-based estimates are projected by the digital twin as described above. These numerically represented image features are normalized and concatenated with those of the digital-twin configuration, then passed through a Multilayer Perceptron (MLP) encoder. The latent embeddings from the ViT and MLP encoders are fused using another MLP with random Dropout and LayerNorm. This layer also enforces the required output dimensionality as the same as the dimension of the parametrized uncorrected configuration $\in \mathbb{R}^{10}$. A hyperbolic tangent function, multiplied element-wise by learnable scaling factors, is used to bound the outputs within joint limits and the endoscope Field of View (FOV), thereby stabilizing model training,

$$\Delta\hat{\theta}^{(i)} = k \cdot \tanh\left[f\left(X_{\text{ref}}^{(i)}, X_{\text{noisy}}^{(i)}, \tilde{\theta}^{(i)}\right)\right], \quad (1)$$

where k is a vector of scaling multipliers, and

$$X_{\text{ref}}^{(i)} = \{\mathbb{M}_{\text{ref}}, \mathbf{p}^{(i)}, \mathbf{l}^{(i)}\}, \quad X_{\text{noisy}}^{(i)} = \{\tilde{\mathbb{M}}, \tilde{\mathbf{p}}^{(i)}, \tilde{\mathbf{l}}^{(i)}\}.$$

The output of (1) is used to correct the noisy estimate $\tilde{\theta}^{(i)}$

$$\hat{\theta}^{(i)} = \tilde{\theta}^{(i)} + \Delta\hat{\theta}^{(i)}, \quad (2)$$

where $\tilde{\theta}^{(i)} \in \mathbb{R}^{10}$ consists of ${}^{\text{CAM}}\tilde{\mathbf{T}}_{\text{PSM}}$ comprising a three-dimensional translation vector and three Euler angles; and the four visible joint angles $\tilde{\theta}_{\text{PSM}}^{(i)}$.

C. Dataset

A research team member teleoperated the dVRK ECM and a PSM equipped with a Large Needle Driver to keep the end-effector in view while producing realistic pose and viewpoint variation. We recorded 100 trajectories with joint positions

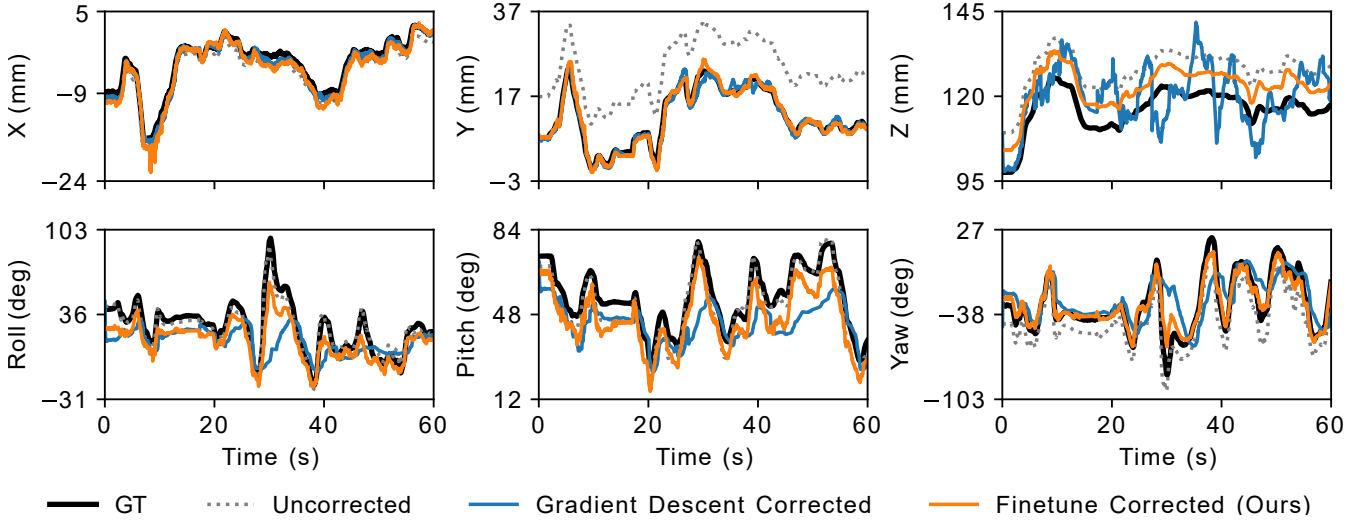


Fig. 3. An example of a trajectory from the test set, showing ground truth, uncorrected configuration, and error-corrected trajectories from the gradient descent approach and our neural network approach corrections, for the end-effector pose.

at 100 Hz and stereo images at 30 Hz; joint positions were downsampled to match the image rate.

Images were rectified and processed with DLC for keypoint extraction and SAM 2 for tool segmentation. Shaft edge lines were then extracted from the segmentation masks using the Hough Line Transform, following Richter et al. [8]. Ground truth end-effector poses were obtained using a MicronTracker 4 (Claronav, Toronto, ON, Canada) optical tracker, while noisy observations were computed from reported ECM/PSM joint angles and a fixed base-to-base transform.

D. Loss Function

After each training forward pass, the digital twin is moved to the corrected configuration $\hat{\theta}^{(i)}$. It then renders the corrected silhouette $\hat{S}$, keypoints $\hat{\mathbf{p}}_{(x,y)}^{(i)}$, and shaft edge lines $\hat{\mathbf{l}}^{(i)}$ with gradient information for backpropagation. The model is trained with a weighted combination of image-space losses,

$$\mathcal{L} = \mathbf{w}^\top \cdot \begin{bmatrix} \mathcal{L}_{\text{mask}} \\ \mathcal{L}_{\text{keypoints}} \\ \mathcal{L}_{\text{shaft}} \end{bmatrix}, \quad (3)$$

where $\mathbf{w}$ balances the numerical scales of the three loss terms. The mask loss aligns the rendered silhouette with the segmented tool mask, including both pixel-wise overlap and silhouette area. The keypoint loss constrains visible end-effector landmarks and remains informative when the silhouette has little overlap with the target mask. The shaft loss follows the polar edge-line representation of Liang et al. [11] and explicitly encourages alignment of the visible tool shaft.

III. RESULTS AND DISCUSSION

A. Tool Tracking

Table I compares the PSM end-effector hand-eye transform ${}^{\text{CAM}}\mathbf{T}_{\text{EEF}}$ error correction performance of our method

TABLE I
TRANSLATION METRICS ON TEST SETS FOR THE PROPOSED METHOD
AND GRADIENT DESCENT.

Method	Metric	X	Y	Z	Norm
Finetune (Ours)	RMSE (mm) ↓	2.49 ±1.15	0.85 ±0.30	6.32 ±0.71	6.33 ±0.70
	NRMSE (%) ↓	9.30 ±4.62	4.95 ±2.40	18.54 ±6.65	18.49 ±6.84
Gradient Descent	RMSE (mm) ↓	2.27 ±1.14	1.00 ±0.26	6.68 ±1.47	6.70 ±1.47
	NRMSE (%) ↓	8.45 ±4.59	6.00 ±2.87	15.70 ±5.94	15.48 ±5.66

against a benchmark technique for translation. The norm vectors were calculated across all axes, and then benchmarked with the ground truth for the norm RMSE.

The Finetune variant denotes our model initialized from pretraining on a synthetic dataset generated by the differentiable digital twin, followed by training on the real robot dataset.

To benchmark our method, we implemented a gradient descent-based method similar to that described in [11], [14]. The method used the same uncorrected configuration and loss functions as our method to iteratively optimize the camera to PSM Roll link transform and the visible four joint angles via Adam. We set a loss convergence threshold equal to our model’s average validation loss, and let it converge on each frame rather than initializing a single correction at the start of a task instance as described in [11]. The maximum number of iterations allowed for a frame was set to 100, which allowed the optimizer to converge to the reference mask within the first one or two frames and consequently propagate its previous prediction as an initialization for its next iteration.

Compared to our approach, the gradient descent method

TABLE II
STEREO INFERENCE PERFORMANCE.

Method	Step (ms) ↓	Iters ↓	FPS ↑
Finetune (Ours)	46.62±0.57	1	21.45±0.25
Gradient Descent	31.67±0.18	11.60±29.62	2.72±6.95

showed slightly lower accuracy, particularly in the y- and z-axes, as shown in Table I. The differences in error between the two methods were within 1 mm. This sub-millimeter difference is likely below the noise floor of the ground truth of the dataset, leading us to conclude that both methods reached the same level of performance. For the selected test set trajectory shown in Fig. 3, both methods produce substantial overlap on the x- and y-axes. However, the gradient descent method had more variability on the z-axis. This was likely due to its error signal oscillating about its convergence threshold.

B. Inference Speed

To evaluate real-time capability, we measured stereo inference speed on a NVIDIA® RTX 3090 GPU with batch size 1. The timing covered the model forward pass and rendering of the uncorrected configuration; the gradient descent baseline was evaluated on the same trajectory with the same acceleration techniques.

As shown in Table II, our method achieved 21.45 Hz with a single deterministic forward pass. Although each gradient descent iteration was faster, the iterative baseline required multiple steps to converge, resulting in approximately 8× lower effective frame rate.

IV. CONCLUSION

In this work, we show a framework that uses the ViT network to estimate the surgical robot end-effector pose using observed image features and uncorrected prediction states. Our experiments demonstrate good performance on the proposed dataset. Also, the method displays a promising real-time processing ability over the gold standard iterative differentiable optimization. This allows it to perform pose corrections online rather than through a one-time calibration at the start of the manipulation, making it less susceptible to drift compared to other differentiable approaches.

In future work, we plan to extend the present translation-axis correction results to full 6D pose refinement.

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